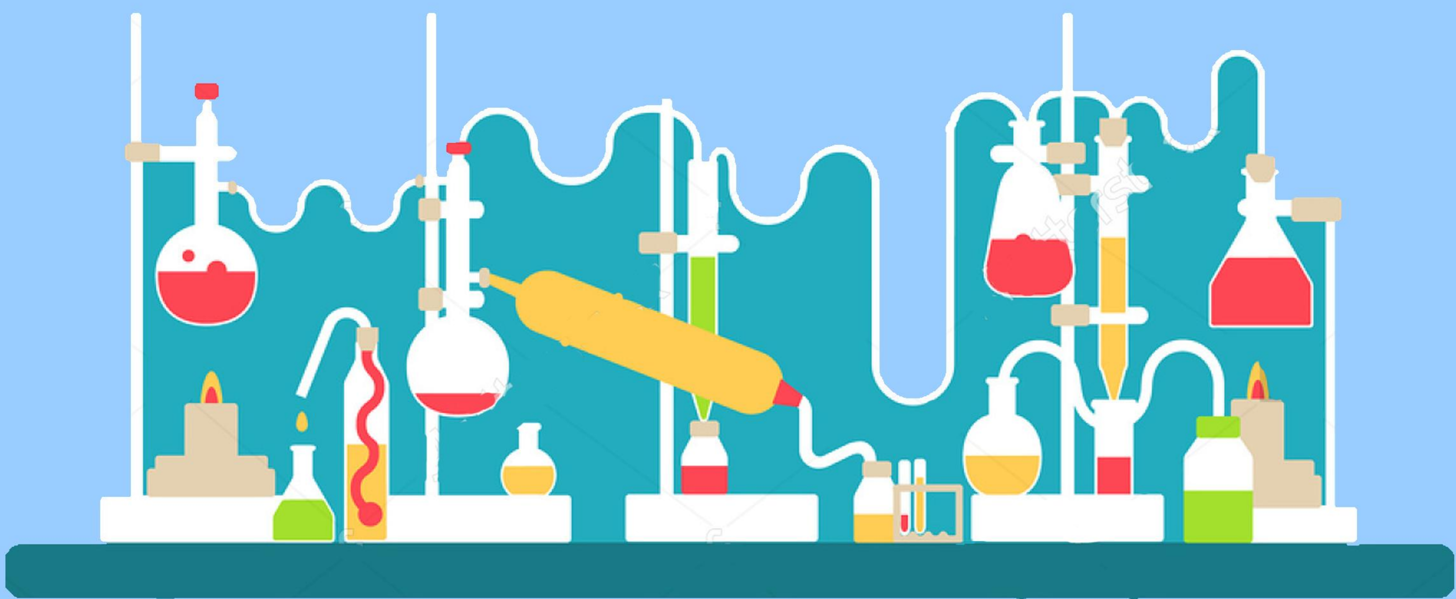
BIOLOGY LABORATORY

لجنة الطب البشري
فكرة تحيا على نبض قلوبكم

لجنة طب الأسنان
Dental Committee
معا نرسم الابتسامة

#1 Microscopes



نضعُ بين أيديكم هذا العملَ الّذي كان نتاجًا للتّعاون بين (لجنة الطبّ البشريّ) و (لجنة طبّ الأسنان)، و نُوجِّهه لطلّاب السّنة الأولى (دُفعة 2016) كمصدرٍ رئيسيٍّ و شاملٍ لمادّة (مختبر البيولوجي) ، و نَضمنُ و فيه شرحٌ لكم اشتمالَه على المطلوبِ منكم من الكتاب و السلايدات، موضِّح للأفكارِ المهمّة، و قد بذلنا جهدنا أن نُخرجه لكم بالرّجوع للمصادر الموثوقة، هذا و قد تجدون بعض التوضيحات باللّغة العربيّة 3: تيسيرًا عليكم؛ إذ هذا أدعى لترسيخ الفهم

تعاون زملاؤكم على إخراجه بصورته المُتقَنة إذ أدركوا مسعى الإتقانِ و مقصدَه، فكان أن وَضعوا أمام أعينِهم صورةَ الطّبيب الّذي ستكونونه كلُّكم، فلم يرضَوا إلّا بأن يكونَ طبيبًا حقًّا، تقديرًا منّا للعلمِ و أهلِه، و إيمانًا بأنّ بذرةَ الخيرِ لا تُنبِتُ إلّا خيرًا :")

كلّ التوفيق ^^"

What is the Microscope?
an optical instrument used for viewing very small objects, it allows us to view structural details of cells, tissues, and organs.

▪ Important concepts related to the microscope:

1- **Magnification**: process of enlarging something only in <u>appearance</u>,not in physical <u>size</u>.
Refers to the "size of an image"

*the magnification of any type lens is engraved on the lens side.

To determine the total magnification use the following formula:

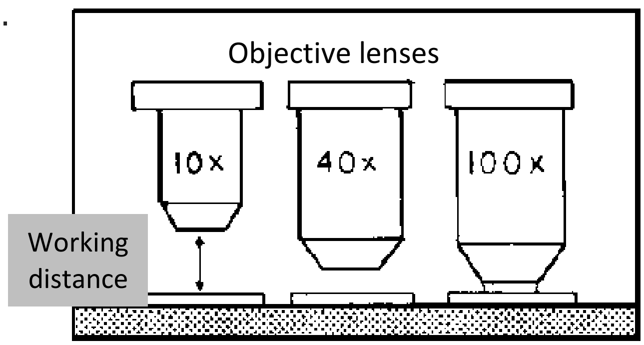
Total Magnification= ocular magnification x objective magnification

نستخدم هذه المعادلة لحساب مقدار <u>التكبير</u> الممكن الوصول اليه باستخدام مجموعة من العدسات،
ستتعرف على هذه العدسات (ocular, objectives) عند دراسة اجزاء المايكروسكوب.

2- **Resolving power (Resolution):** the ability of the microscope to separate or distinguish small or closely adjacent images. It refers to the "clarity of an image".
(It depends on the light source)

3- **Working distance:** is the distance between the stage the objective lens.
The objective lenses vary in their lengths, so the working distance will change as you change from one objective lens to the next.

مسافة العمل هي المسافة بين المنصة الشريحة والعدسة وهذه المسافة التي يمكن ان تتحرك فيها الشريحة اقتراباً او ابتعداً عن العدسة. تقل هذه المسافة كلما زادت قوة تكبير العدسة (magnification)



As the lens magnification increases,
The working distance Decreases.

4- **Contrast:** the ability to show unlikeness or differences; the opposite to the background.

5- **Diameter of field of view:** is a characteristic of the microscope that allows us to calculate the approximate size of the specimen seen with microscope in different conditions.
If we have the magnification & DFV for one objective lens, we can calculate the DFV for another objective lens on the same microscope.
**** this concept can be applied only with the (Parfocal Microscopes)**
(it depends on the magnification, as it increases the DVF decreases.)

M1 x DVF1 = M2 x DFV2

- **Microscopes types:**
  Microscopes can be classified based on the physical principle that is used to generate an image; Microscopes visualize different physical characteristics of the sample for example:
  -Elasticity of the sample
  -Image contrast
  -Resolution
  -Destructive of the samples

1- Optical microscope: this type of microscopes uses visible light OR UV light

This type of microscopes includes:

Compound M	Stereo M (Dissecting M)	Dark-field M	Phase contrast M	Fluorescent M
Visible light	Visible light	Visible light	Visible light	UV light
2D	3D	2D		
Maximum Magnification 1000x	Maximum Magnification 100x			
Have two lenses system: Objective and ocular -produces true inverted image	يستخدم مع العينات غير الشفافة opaque objects & large specimens. -produces non-inverted image -have two light systems: transmitted & reflected. -large working distance.	-Light object is seen on dark background -Used with living spirochetes like syphilis pathogen	-Have special condensers that throw light through the object in different speeds -Used to see living unstained organisms &internal parts of the cell.	-Use anti-bodies technique that involve: fluorescent dyes & antibodies -UV light excites the electrons of the objects so they give off light in various colors.

2- X-ray Microscopes:
   -use X-ray beam to create image.
   -create higher resolution image due to the small wave length of X-ray than light microscopes.
   -have higher magnification that light microscopes but less than electron microscopes.
   -used to observe living organisms

3- Scanning acoustic microscope (SAM):
-use focused sound waves to create image.
-used to uncover elasticity, stress, tension inside biological structure.
-used to detect small cracks or tensions in material.

4- Scanning helium ion microscope (SHIM or HeIM)
-uses beam of He ions to generate an image.
-leave the sample intact سليم unlike the Electron M; because He ion beam has low energy unlike the accelerated electrons.
-high resolution image

5- Neutron microscope:
-high resolution
- the best contrast in all microscopes.

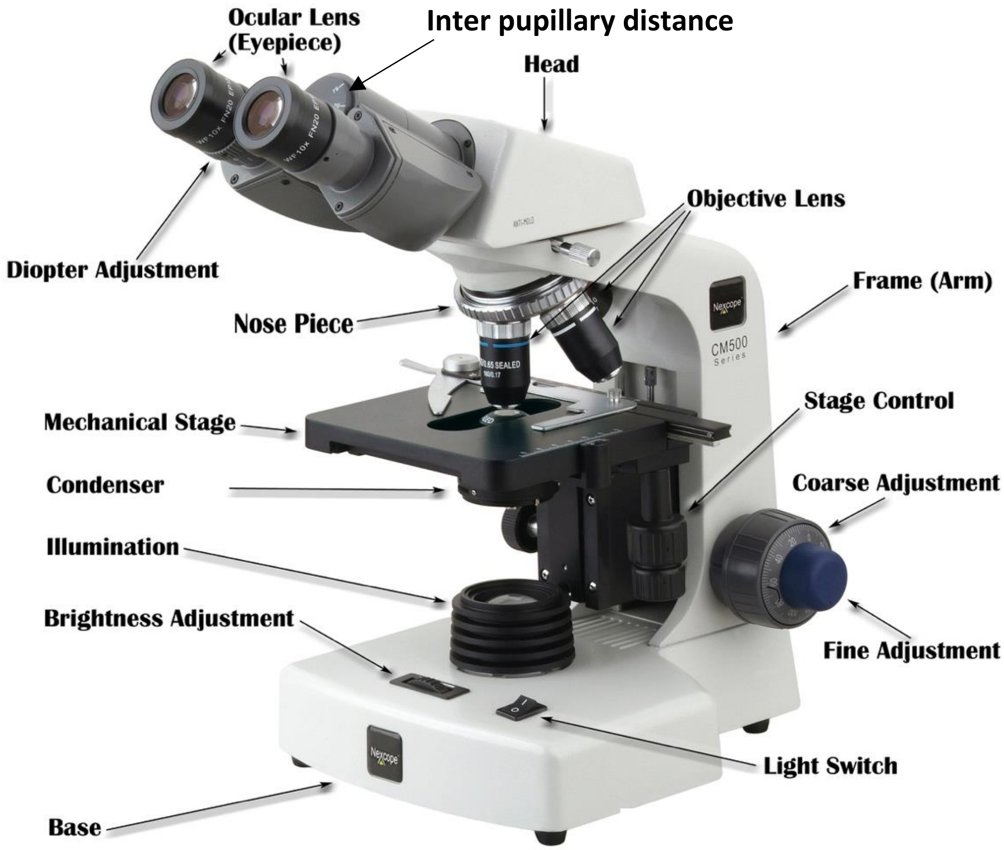
6- Electron microscope:
-magnification to 2 million time (2000000000x)
-uses high energy accelerated electrons with thin layer of metal coating
-very tough on the sample (high destructive)

there are two types of Electron microscopes:

Transmission EM	Scanning EM
2D	3D
Electrons pass through the sample	Electrons bounce off the sample

7- Scanning probe microscope:
-used to generate images for atoms

Light/student Microscope parts:



Let's see each part and its function:

-The microscopes common used in our lab are binocular or monocular light microscope.
Binocular microscopes have two eyepieces. Monocular microscopes have one eyepiece.

-They are called compound microscopes because a specimen is viewed through a series of lenses.

1-The eyepiece of the ocular lens(عدسة العين): is located on the top of the microscope. It normally has magnification of 10x (10x ثابتة و تساوي دائما ocular قيمة تكبير عدسات ال)

so the total magnification for will be = 10 (ocular magnification) x 4/10/40/100 (objective magnification)

-Be careful these lenses are not attached, and therefore may fall down unless the microscope is kept upright.

2-Body tube: Carry the ocular lenses and joins them with nosepiece.

3-The resolving nosepiece: is the revolving part to which the objective lenses are attached, uses to rotate the objectives in positions.

4-Objectives are lenses: located on the moveable resolving nosepiece. They usually have different color strips.

name	length	magnification	Working distance
The scanning lens	shortest	4x	greatest
The low-power lens	Longer than the 4x	10x	Second greatest
The high-power lens (high-dry lens)	Longer than the 4x and 10x	40x	Less working distance than the 10x and 4x
The oil immersion lens	The longest	100x	The shortest working distance

- We only use oil immersion lens with immersion oil. While the others are used with H_2O

5-Stage: Support the slide that is held to it by two clips, and has a hole so that light can be pass through the specimen.

6-Fine focus adjustment knob: makes final or fine focusing on a specimen. It is used with oil immersion lens (100X), and with high-dry objective (40 X) if there is image not clear.

7- Coarse focus adjustment knob: brings the specimen in focus under the scanning and low-power objectives (4X and 10X) and with high-dry objective (40X) if there is no image at all.

8-Stage adjustment knob (mechanical stage control and adobe knobs):
- control the movement of stage, therefore the movement of slides.

9-The arm: is the curved of the frame.
- It supports upper parts of a microscope and provides a carrying handle.

8-The condenser lens: is to concentrate and direct the light.
There is an adjust knob to control that height of the condenser.

9-The iris diaphragm(CLIP): controls the amount of light passing through of the specimen. It is controlled by a lever(clip) that move back and forth.

10-The base: acts as a stand for the microscope and houses the lamp.

11-Light switch: turn on/off the illuminator of the microscope.

12-Light source: usually a small electric light beneath the stage that controlled by light switch.

Important not to keep in mind

Light microscope image characteristics: TRUE INVERTED image

-When the stage moves to the right so image in the lens moves to the left, and vice versa.
-when the stage moves downwards so the image in the lens gets bigger, and vice versa.

قبل البدء في الحديث عن كيفية استخدام المجهر، لا بد من الالتزام ببعض القوانين في التعامل مع المجهر من اجل سلامة المجهر والسلامة الشخصية:

1 - احمل المجهر بكلتا يديك، بحيث تضع احداهما اسفل قاعدة المجهر baseو الاخرى على ذراع المجهر arm.
2 - تاكد ان الطاولة مرتبة و خالية من الفوضى قبل ان تضع المجهر.
3 - لا تجر المجهر على الطاولة، اذا اردت تحريكه فعليك ان تحمله و تنقل الى حيث تريد.
4 - عندما تضع المجهر على الطاولة، اجعل عدسات ال ocular دائما باتجاهك.
5 - يجب ان تكون حافة قاعدة المجهر بعيدة على الاقل ٥ سم او ٢ انش عن حافة الطاولة
6 - لا تشغل المجهر الا عند الحاجة اليه فقط
7 - ازل عن المجهر الغطاء و نظفه باستخدام Lens Paper.

(((كيف نستخدم المجهر))))
هذه الخطوات كلها للفهم

1. Clean the microscope with lens paper, plug in the electrical source, and switch on the light source.
2. Use the **coarse focus adjustment knob** to maximize the **working distance** (the distance between the stage and the objective lens), this means: lower the stage to the lowest point.
3. Rotate the revolving **nosepiece** into position with the **scanning power (4X) objective** lens in the viewing position.
4. Center the slide holder of the mechanical stage on the microscope stage.
5. Place the slide between the clip and push it all the way back to the bar.
6. Using the **mechanical stage drive knobs**, center the coverslip and specimen over the stage aperture.
7. While carefully watching the slide on the stage, use the **coarse focus adjustment knob** to move the specimen towards the **scanning objective lens** until it stops (raise the stage). The stage will come close to the lens but will not touch it.
8. Adjust the **interpapillary distance** until you see a single circle while looking through the microscope with both eyes open. This circle of light is called the **field of view**.
9. While looking through the **ocular lenses**, turn the **coarse and fine focus adjustment knobs** of the microscope until you see something you believe is the specimen. **Stop**. Move the **slide** back forth using the **mechanical stage drive knobs** the item you thought was specimen should likewide be oving back and forth.
10. Cover of close the eye that is not looking through the ocular containing the **diopter ring**. Viewing with only that eye focus using the **coarse focus adjustment knobs**. Adjust the light using the **iris diaphragm adjustment** lever and/or the **light adjustment**. Then close your other eye adjusting the **diopter ring** on that ocular lens to bring the object into focus.
11. Adjust the **condenser** to the highest position.
12. Using the **mechanical stage drive knobs**, center the specimen of choice in the viewing area.
13. These microscopes are **parfocal** (if one lens is in focus, all other lenses are, at least, close to focus). In order to change to the next highest magnification, simply rotate the **nosepiece** to the **low power (10X) objective lens**.
14. These microscopes are also **parcentral** (if an object is in the center of the field of view for one lens, it will be, at least, close to the center of the field of view at other lenses).
15. Using the **mechanical stage drive knobs**, re-center the specimen in the viewing area.
16. With the **low power (10X) objective**, use the **coarse focus adjustment knob** to focus the view of the specimen and the **iris diaphragm adjustment** lever to increase the light intensity on the specimen. Re-center the specimen in the field of view. Rotate the **nosepiece** to the **high power (40X) objective lens**. If there is:
    Image but not clear use the **FINE FOCUS ADJUSTMENT KNOB ONLY** to focus the specimen.
    No image use **COARSE FOCUS ADJUSTMENT KNOB** to focus the specimen.
    Then use the iris diaphragm adjustment lever to increase the light intensity on the specimen. If needed, use the light adjustment to provide additional light.

17.Re-center the specimen the specimen in the field of view. Rotate the nosepiece so no objective lens at the position. Add a drop of oil immersion on the slide at light spot region. Then rotate the nosepiece to oil immersion lens (100X).

17. Use the **FINE FOCUS ADJUSTMENT KNOB** only to focus the specimen.
18. When removing the slide, rotate the nosepiece so the scanning power (4X) objective is in the viewing position, then use the **coarse focus adjustment knob** to maximize the **working distance (lower the stage)**.
19. After you have completed the laboratory activity, turn the light switch off. Clean all microscope lenses (objective and ocular) with **lens cleaner** and **lens paper**.
20. Prepare the microscope for **storage** using the checklist below. Be sure:
    - A. the **scanning power (4X) objective** is in the viewing position.
    - B. The **mechanical stage** has been positioned so the stage arm is flush with the right side of the stage.
    - C. The cord is wrapped securely around the microscope arm.
    - D. The **stage** has been adjusted all the way down.

فقط للفهم، ربتها بعقلك ، افهم الهدف من كل خطوة لا اكثر و بس تجرب بيديك هذه الامور ستحد كم هي سهلة و جميلة ^^

Test your self:

Q1)Choose the the correct answer:

1. **The lens at the top that we look through are?**
   a) Objective B)ocular C)filter lenses D)tube
2. **The total magnification power produced by a microscope using 10x objective lens is ?**
   a) 40x B) 1000x C)400x D)100x
3. **always begin examining microscope slides with which power objective?**
   a) scanning B)Low C)High D) Oil immersion
4. **at high power , always use which part of the microscope to focus the image?**
   a) Coarse B)condenser C) fine D)clips
5. **The magnification of ocular lens:**
   a) 100x B)12x C)10x D)20x

Q2) true or false.

1. **Always hold the microscope by the arm and the base _______**
2. **Make sure to use the high power lens first ______**
3. **Use your soft shirt to clean the lenses or light source ______**
4. **Make sure to place the microscope away from the edge the table ___**
5. **It is okay to place random objects on the microscope without slide ____**

Key answer

Q1 1b 2d 3a 4c 5c

Q2 1t 2f 3f 4t 5f